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Project Proposal Report

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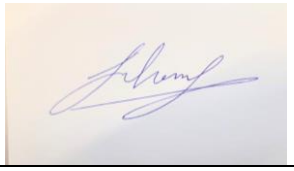
Department of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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ABSTRACT

The disaster relief camps are facing a significant challenge in managing the identification of humans, distribution of resources, and communication with severely limited infrastructures. The existing IoT-based disaster management systems are found to be heavily dependent on the availability of Internet connectivity and are providing equal communication priority for all messages sent by the system, thereby resulting in severe data loss during communication. This is leading to a significant risk for disaster relief camps, as the loss of biometric identification data and resource allocation information may directly affect the safety of humans.

The proposed research aims to introduce a Smart Camp Box, which is a portable IoT system that can be used for disaster relief camps with severely limited Internet connectivity. The proposed system includes a pre-disaster fingerprint registration system and a wristband-based barcode identification system, thereby providing a reliable system for identifying humans and managing resource entitlement without the need for Internet connectivity. The people are registered with their biometric data before the disaster strikes, and upon arrival at the camps, people are identified through fingerprint scanning and provided with uniquely linked barcodes that manage resource allocation, thereby avoiding duplication and duplication of resources.

The technical contribution of the proposed research is the development of a Semantic-Aware Adaptive Message Prioritization framework for MQTT communication protocols. Unlike existing techniques that use static quality of service configurations and message content keywords for message prioritization, the proposed framework categorizes all system messages based on the source type and action that triggers them and assigns them with Critical, Important, and Routine levels of criticality. A real-time network state monitor continuously monitors the latency and packet loss rate of the network and determines the quality of the connectivity. The proposed framework dynamically configures the quality of service levels for MQTT communication based on the combined evaluation of message criticality and network state, thereby ensuring that critical messages like biometric registrations and distribution confirmations are always given priority and buffered for guaranteed delivery upon reconnection.

The proposed framework is compared with static MQTT configurations for communication in a simulated environment with network degradation scenarios, and the results show that the proposed framework improves communication reliability in disaster IoT environments with constrained networks.

Keywords: Internet of Things (IoT), Location-Aware Disaster Monitoring, Flood and Landslide Early Warning Systems, GPS-Based Environmental Monitoring, Relief Camp Risk Assessment, .

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LIST OF ABBREVIATIONS

Abbreviation	Description
IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
QoS	Quality of Service
SACS	Source-Action Classification Schema
DSR	Design Science Research
UNDP	United Nations Development Programme
UNHCR	United Nations High Commissioner for Refugees
DMC	Disaster Management Centre
NDRSC	National Disaster Relief Services Centre
API	Application Programming Interface
HTTP	Hypertext Transfer Protocol
Wi-Fi	Wireless Fidelity
ML	Machine Learning

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1. INTRODUCTION

1.1 Background and Context

The Real-World Problem and Who It Affects

Natural disasters pose one of the most critical humanitarian crises of the modern world, which requires swift and well-coordinated responses on an extensive scale and in an environment of critical infrastructure collapse. Among the countries that have faced the brunt of natural disasters on an ongoing and consistent scale, Sri Lanka presents an interesting and critical case. Between 2010 and 2025, natural disasters in Sri Lanka have impacted more than 15.5 million people, displaced around 1.5 million people, and resulted in the death of 1,328 people, with floods and landslides being the most common and impactful type of natural disaster. The compounding frequency of these events, which is being driven by the monsoonal climate cycles and the increase in climate variability that the country is experiencing, has revealed the structural vulnerabilities that the national disaster responses have in the realm of camp management and the identification of humans and the distribution of resources.

The most devastating recent manifestation of the structural vulnerabilities that the national disaster responses of the country have was the impact of the Cyclonic Storm Ditwah that made landfall on the eastern coast of Sri Lanka on the 28th of November in the year 2025. The impact of the cyclone was devastating as it resulted in the displacement of more than 1.4 million people across the entire 25 districts of the island nation. The cyclone resulted in the deaths of 410 people and also caused the disappearance of 336 people. Nearly 233,000 people were displaced due to the extensive damage that the cyclone caused to homes and infrastructure.

The magnitude of the disaster put an overwhelming burden on the national relief infrastructure. At the height of the disaster, there were 956 active relief centers across Sri Lanka, sheltering a total of 27,663 families and 89,857 people, with over 275,000 children affected by the disaster.

Who is affected and why it is important

The populations that are most affected by disaster events in Sri Lanka are those that are less able to recover from them. From the analysis of data by the UNDP and the University of Oxford, it was established that 48 percent of Sri Lankan households lacked the capacity to adapt to disaster events when Cyclone Ditwah struck, adding to existing risks from Sri Lanka's worst economic crisis in 2022. Within the relief camps, the displaced people, including children and the aged and those suffering from chronic health conditions, are entirely dependent on the relief camp administrations for food, water, medicine, and shelter. The reliability of the identity and resource allocation systems in the relief camps is what will either provide relief to the displaced people or cause them to languish without relief due to administrative failures, duplications, and corruptions.

The children in the relief camps are a particularly vulnerable segment of the displaced people. UNICEF personnel operating during the relief efforts for the victims of the Cyclone Ditwah worked tirelessly to deliver portable water to the safety centers and educational materials. The absence of structured, technology-supported camp management systems means that even these basic welfare activities are conducted through manual, error-prone processes that scale poorly under mass displacement conditions.

Statistical evidence (local/global)

Sri Lanka is ranked at 104 out of 191 countries according to the 2024 INFORM Risk Index. The country has an estimated 87 percent of its population living under areas of medium or severe climate impact exposure. The country is witnessing a drastic increase in climate-related disasters. The country witnessed 304 climate-related disasters from 2015 to 2022, compared to only 35 from 2008 to 2014. The trend is only moving upward, making it necessary to invest in scalable technology-enabled infrastructure for disaster response.

In October 2024, a total of 155,000 people from the North Western, Sabaragamuwa, and Western provinces of Sri Lanka were affected by intermonsoon flooding. The affected population needs assistance from 40,500 people. The needs are pressing, mainly for food, drinking water, sanitation, and healthcare. The country is witnessing a recurring crisis that is exposing a lack of capabilities to identify affected people accurately, a lack of a reliable system to prevent duplication of resources distribution, and a lack of adequate communication infrastructure during disasters.

Analysis of existing solutions and their limitations

The existing system of managing disaster relief camps in Sri Lanka, under the administration of the National Disaster Relief Services Centre (NDRSC), is greatly dependent on manual registration systems and resource distribution records, as well as oral coordination between administrators in these camps. The technological response to Cyclone Ditwah highlighted the critical weaknesses in this system in a dramatic fashion. There were reports of power outages, major road blockages, and communication network disruptions in all high-risk zones, including Batticaloa, Ampara, Puttalam, Gampaha, Kandy, and Colombo. These were essentially zones isolated from any sort of relief operations. There was no proper coordination between major institutions such as the Disaster Management Centre, National Disaster Relief Services Centre, and National Building Research Organisation during the Cyclone Ditwah relief operations. The authorities did not communicate critical information to the public in all three national languages as required by policy. This lack of institutional coordination was further complicated by the breakdown in digital communication networks exactly when these were required. There is no existing IoT-based disaster management system that is built to function as an offline-first, edge computing system that is capable of maintaining full operational integrity without the presence of an internet connection exactly the situation that characterises the most critical phases of any given disaster event in Sri Lanka. Moreover, existing communication frameworks for IoT systems fail to address the heterogeneity in criticality associated with messages transmitted from within a disaster relief camp. For instance, a message confirming that a displaced person has been provided with their daily quota of food has significantly different criticality if lost under network conditions compared to a system “heartbeat,” yet existing MQTT-based systems treat these messages with equivalent criticality under degraded network conditions. This creates a critical failure condition in that critical identity and distribution information are given equal priority to operational data under bandlimited conditions.

Ideas for the proposed solution and its parts

The research presents the concept of a Smart Camp Box, a portable self contained IoT solution intended for use in disaster relief camps under the infrastructure conditions common in Sri Lanka. It creates a self contained WiFi network at the point of deployment, removing the need for an external internet connection in the operation of the core management activities in the disaster relief camps. Disaster victims are biometrically registered using a fingerprint scanning process prior to the disaster event. On arrival at the disaster relief camps, the verified victims are provided with a uniquely linked barcode wristband that acts as their identification token in all subsequent interactions with resource allocation activities.

The main technical contribution of Smart Camp Box is that it employs a Semantic-Aware Adaptive Message Prioritization framework for all internal IoT communication using MQTT. This framework continuously monitors real-time network quality parameters such as latency and packet loss rate, categorizes all system-generated messages based on source type and action that triggers the message into three levels of criticality – Critical, Important, and Routine – and determines MQTT Quality of Service and priority based on a combined evaluation of message criticality and network state. During a disconnection, messages are buffered locally and sent out in strict priority order upon reconnection, ensuring that biometric registration and resource distribution confirmations are never lost.

The system also includes a cloud-based machine learning prediction module that uses historical data related to the registration and distribution of resources to predict future resource demand, thereby helping the camp administration manage resources proactively. Educational content and awareness material are served locally using HTTP and WebSocket protocols for displaced children and the community at large, without the need for an internet connection.

Relevant SDG alignment

This research directly contributes to the achievement of various United Nations' Sustainable Development Goals. Specifically,

SDG 1 No Poverty is addressed by ensuring that resources are distributed reliably to ensure that entitled support is provided to all displaced individuals despite connectivity conditions.

SDG 3 Good Health and Well-Being is addressed by ensuring that medical request messages are given priority to ensure that health needs are delivered to administrators without loss.

SDG 4 Quality Education is addressed by ensuring that the offline subsystem for delivering educational content to displaced children within refugee camps is operational.

SDG 9 Industry, Innovation, and Infrastructure is advanced by this research through the novel IoT communication framework proposed.

SDG 11 Sustainable Cities and Communities is directly addressed by enhancing resilience within disaster response infrastructure.

SDG 13 Climate Action is addressed by enhancing resilience within communities to respond to various climate related disasters that Sri Lanka is increasingly experiencing.

Research cluster alignment

The research intersects with three active clusters of academic research. The first cluster is the Internet of Things and Edge Computing cluster, with the sub-cluster being Constrained IoT Communications

in Infrastructure Degraded Environments. The second cluster is the Humanitarian Technology and Disaster Informatics cluster, which deals with technology design for humanitarian operations. The third cluster is the Adaptive Communication Protocols cluster. It includes the sub-clusters of quality of service management, message prioritization, and optimization. The intersection of these three clusters is a relatively uncharted area of research that the proposed work targets.

1.2 Literature Review

2.1 Overview

The following literature review covers three interrelated research domains, all of which, in combination, creates the foundation on which this research is based. These domains include IoT-based communication protocols and Quality of Service, message prioritization in MQTT protocols, and biometric identity management in humanitarian and disaster scenarios. Twelve scholarly pieces published between 2018 and 2025 have been reviewed, including existing research methodologies, existing products and systems, and those that create the existing research gap.

2.2 IoT Communication Protocols and QoS Management

2.2.1 MQTT as the Foundation Protocol

MQTT is a lightweight, publish-subscribe, machine-to-machine network protocol primarily designed for connections to remote locations containing constrained devices. Due to the above characteristics of the MQTT communication protocol, it is the most widely used communication protocol in IoT applications. It queues messages and retries them when necessary. It is able to function without the need for an internet connection by using a local area network with a local broker. These characteristics are critical for the implementation of the industrial IoT communication protocol. However, despite the above advantages of the MQTT communication protocol, the standard MQTT communication protocol is limited in the context of a disaster. Although the MQTT communication protocol provides three levels of quality of service for different classes of traffic, the flow of the traffic from the publishers to the subscribers is not controlled. The publishers send the data to the broker, and the broker sends the data to the subscribers in the same order. The lack of a reliable flow control mechanism increases the probability of packet losses. This means that under bandwidth degradation, all messages regardless of their operational importance compete equally for transmission resources a dangerous phenomenon in a disaster relief scenario where the criticality of messages varies greatly from system components. Limitation Identified: The standard MQTT protocol lacks a native capability to differentiate the importance of messages or protect messages of higher criticality during bandwidth degradation.

2.2.2 Adaptive QoS for MQTT in Constrained Networks

In light of the static nature of the QoS in MQTT, various dynamic QoS management approaches have been proposed. Palomero et al. (2022) propose a dynamic QoS controller based on the MQTT-SN protocol. The dynamic QoS controller assesses the status of the network in terms of the end-to-end delay and packet error rate of the network. It reacts accordingly by automatically assigning the optimal QoS value to each node in the network. It optimises the quality of service at the application layer based on an analysis of simple and easy-to-compute performance indicators derived from the network in use. Although not a direct match with the network monitoring component of the proposed research, it shows the feasibility of adapting QoS in a network based on various network metrics.

However, this work's applicability to disaster relief contexts is restricted by a number of important limitations. The controller operates in a simulated NS-3 environment instead of actual deployment conditions, is constructed for MQTT-SN instead of standard MQTT, and most importantly applies a consistent QoS adjustment across all topics without taking message semantic content or source type into account. A biometric registration record and a heartbeat status update would be treated the same under the same network conditions because all messages in the system receive the same QoS level based only on network metrics.

This restriction is partially addressed in the context of Electric IoT systems by Zhang et al. (2022). They conclude that MQTT QoS levels for data transmission must be dynamically and intelligently chosen in accordance with time-varying network state and QoS requirements. They note that QoS 0 produces relatively lower transmission delay but higher packet loss ratio, while QoS 1 and QoS 2 achieve no packet loss at the expense of increased transmission delay. They also note that although the QoS requirements for acquisition and control services differ in terms of reliability and delay, these disparate metrics are inconsistent. Crucially, the same paper clearly identifies the research gap: the literature right now in publication has not sufficiently addressed joint optimization of delay and packet loss ratio under uncertain real-world network conditions. The issue this study aims to address is directly validated by this published acknowledgement of the gap.

Limitation noted: Current adaptive QoS research either focuses on industrial/electrical IoT domains with no relevance to humanitarian disaster relief camp environments, or it applies network-aware adaptation without message classification.

2.3.2 Real-Time and SDN-Based Priority Extensions

Shahri, Pedreiras, and Almeida (2022) extend MQTT with real-time communication services using Software Defined Networking. Their RT-MQTT system introduces a priority control mechanism for heterogeneous remote monitoring IoT systems where a Priority Broker controls communication link utilization, achieving latency reductions and increased successful message delivery proportional to message priority assignment. While technically sophisticated, this approach requires dedicated SDN infrastructure as a requirement entirely incompatible with disaster relief camp deployment conditions where no preexisting network infrastructure can be assumed. The approach also evaluates performance exclusively under stable connectivity conditions, providing no evidence of behavior under intermittent or degraded network states.

Uchida et al. (2022) apply MQTT priority controls to an early landslide warning system, demonstrating domain-specific priority assignment for disaster adjacent applications. While this work establishes precedent for MQTT prioritization in natural disaster contexts, it focuses on sensor data transmission for warning systems rather than the camp management, human identification, and resource distribution functions that characterize the Smart Camp Box use case.

Limitation identified: Advanced MQTT priority extensions require infrastructure unavailable in disaster deployments, and domain-specific applications focus on environmental sensing rather than camp management functions.

1.3 Research Gap

1.3.1 What Current Research Is Missing

Five unique and related gaps are identified by a systematic review of the literature in three domains: MQTT communication and QoS management, message prioritization in IoT systems, and IoT applications in emergency and disaster contexts. These gaps together define the research problem that this study attempts to solve. Direct evidence from published literature is used to support each gap.

Gap 1 Lack of an Existing System that Simultaneously Carries Out Message Priority Classification and RealTime QoS Adaptation

The first gap that has been recognized is the lack of an existing system that has the capability to simultaneously perform the tasks of message priority classification as well as the adaptation of the QoS level based on the real-time network conditions.

In the research done by Akshatha and Dilip Kumar (2024), the authors show that the queuing of priority-based messages helps in improving the delivery of urgent messages. However, the classification is static in nature. The classification is done based on the topic labels. On the contrary, the demonstration of Palomero et al. (2022) shows how the adaptive quality of service controller adjusts the quality of service according to the network latency and error rate, but with the same quality of service for all messages, regardless of their operational significance. The quality of service for a heartbeat status message is the same as for a biometric registration record under the same network degradation.

This limitation is explicitly stated in the industrial IoT literature, which argues that the quality of service levels for MQTT do not satisfy the requirements for the constrained environment, as the losses of messages with quality of service level 0 are not acceptable for the messages with critical information, while the quality of service levels 1 and 2 cause the queues, which can lead to catastrophic failures due to the rapid change of the values of the messages, with the quality of service level being a difficult compromise that cannot be made by the MQTT protocol itself.

Authors Zhang, et al. (2022), whose research is the most relevant to the domain of adaptive QoS, explicitly state within their conclusions, which are readily available, that the issue of the joint optimization of delay and packet loss ratio under uncertain network environments was not adequately addressed. This represents direct evidence of the existence of the gap from within the research community.

What is missing: A unifying decision mechanism that considers both what the message is, as well as what the network is, with the goal of achieving optimal QoS because of the evaluation.

Gap 2 No existing system classifies message criticality based on source type and triggering action

All existing MQTT message prioritization schemes that we are aware of, including Priority-enabled MQTT (2024), PrioMQTT (2024), PR-MQTT (2024), and Puthiyidam and Joseph's (2022) prioritization survey, classify messages based on labels of message topics, content of messages, or message type identifiers. None of these schemes classify messages based on the combination of source type and triggering action of the message.

This is an important distinction in the context of a disaster relief camp scenario. The source of a message, be it a fingerprint biometric sensor, a wristband barcode reader, a web form input, or a system monitor, carries inherent information that is related to the criticality of that message that cannot be represented by any other means than through its source type. A message received from a wristband barcode reader indicating that resources are being distributed is a legal entitlement that cannot be recovered if lost. A message from the system monitor regarding storage capacity does not have such an effect. Though the QoS mechanism of MQTT enables programmers to choose appropriate QoS based on their application needs, no such mechanism is defined for the automatic inference of criticality; it is entirely up to the programmer to decide it, which is done statically during design time.

No existing published work has proposed a source of action classification schema for MQTT messages for any application domain, including the domain of managing disaster relief camps. This is a new classification mechanism not found in existing literature.

What is missing: A formally defined source-type-driven semantic message classification schema that can automatically infer message criticality based on context rather than content or manual assignment.

Gap 3 No Existing IoT System Implements Offline First Operation with Intelligent Message Buffering and Reconnection Flushing of Queues in a Priority Order

Published technical analysis of existing IoT protocols, like MQTT, revealed that it has inherent disadvantages in complex network environments, especially when implementing a TCP-based transport protocol, such as those experienced with IoT devices, where connections are frequently interrupted during network switching, it is difficult to re-establish a network connection after disconnection, and blockage of data transmission occurs due to network congestion, loss, and retransmission in poor network environments. Despite these disadvantages, no existing IoT system was found that provides a comprehensive model for implementing an offline-first operation, where intelligent message buffering occurs during network disconnection and flushing of these messages in a priority order when a network connection is re-established.

The standard mechanisms for implementing persistent sessions in MQTT allow for the storage of unacknowledged QoS 1 messages for a disassociated client that will be sent on reassociation up to a rate of 10 messages per second. However, the messages will be sent in the order that they were received without any priority ordering. Additionally, any messages that are sent that exceed the account storage limits will be discarded without any selection based on the level of criticality. This creates a situation in a disaster camp in which the order that the messages are sent to the cloud after a long-disassociated time is completely dependent on the order that they received. This is a dangerous situation since a backlog of messages may contain registration data as well as more mundane heartbeat pings.

1.3.2 Why Current Approaches Are Insufficient

Research has already proven that infrastructures such as mobile networks and radio towers may be overloaded, damaged, or even completely inoperable during major disaster events, causing a breakdown in coordination that exists within these systems. The existing systems do not have strong resilience against disruptions or overloads in networks during disasters. The standard implementation of MQTT is not designed to address this reality. The QoS levels are implemented when connectivity is present but do not offer any intelligent response when it is not present.

The implication of this insufficiency in a disaster relief camp scenario is that the performance of the system is not the issue: it is the loss of the records that determine who gets food, water, and medicine. When the bandwidth is constrained and a standard MQTT broker is used to handle all messages with the same priority, the loss of a biometric registration message is statistically equivalent to the loss of a routine heartbeat message. The system has no way of knowing that one loss is catastrophic, and the other is completely irrelevant.

1.3.3 Justification of Novelty and Creativity

The novelty of the proposed framework of Smart Camp Box is based on three pillars that are individually not present in existing literature but together create something that is not present in any of the existing literature. Pillar 1 – Methodological Novelty: The proposed Source-Action Classification Schema (SACS) proposes a novel approach for determining the criticality of messages in IoT systems based on the source of the message and the action that led to its transmission, instead of relying on content analysis or manual assignment of topics, resulting in a classification that is not only automated but also resilient against changes in message format and is based on the semantics of the specific system under consideration, which has not been proposed or evaluated in any of the existing literature.

Pillar 2 - Technical Novelty The Semantic-Aware Adaptive QoS Management framework is the first published system to combine the runtime network state monitoring and semantic message classification functions into a single QoS assignment algorithm. The decision matrix formed by the combination of message criticality level and network state to an optimal quality of service assignment/routing decision represents a formally specifiable algorithm whose output can be shown to be measurably distinct from both static quality of service assignment and network state-based quality of service adaptation algorithms under the disaster network degradation scenario.

Pillar 3 - Domain Novelty The application of the offline-first edge computing IoT architecture to the disaster relief camp management problem space, including pre-disaster biometric registration, wristband-based resource entitlement, and adaptive MQTT communication, represents a problem space entirely absent from the existing body of published works on IoT communication systems and disaster response technologies. The combination of the disaster risk profile of Sri Lanka, the operational needs of displaced persons management, and the technical limitations of edge computing-based IoT systems represents a problem space that is not only practically urgent but also entirely unexplored.

- Clearly state what is missing
- Why current approaches are insufficient
- Justify novelty and creativity
- Evidence-based argument

Research Gap	Evidence
No combined priority + adaptive QoS joint optimization	Zhang et al. (2022) explicitly stats of delay and packet loss is unaddressed
No source-type classification schema labels or content combination	All reviewed papers use static topic keywords none use source + action
No offline-first priority buffering arrival order with IoT, 2024)	MQTT persistent sessions deliver in no criticality-based preservation (AWS
No offline biometric humanitarian IoT require	UNHCR BIMS and all institutional systems

Each of these gaps is individually justified through evidence provided in existing literature, peer reviewed surveys, or institutional documents. The convergence of all these gaps into a single deployment scenario of the offline first disaster relief camp IoT system is the research problem that this framework of Smart Camp Box is intended to address, which is also the novelty of this research contribution to the existing literature.

2. OBJECTIVE

2.1 Main Objective

The design, implementation, and evaluation of a Semantic Aware Adaptive Message Prioritization framework for an offline-first IoT system for disaster relief camp management with guaranteed reliable and prioritized communication of critical operational data under low bandwidth and intermittent internet connectivity conditions, based on the implementation of a prototype for the context of disaster relief scenarios in Sri Lanka.

2.2 Specific Objectives

Objective 1

Develop a pre-disaster biometric registration and real-time identity verification system based on fingerprint scanning with wristband-based barcode identification for the management of resources within the disaster relief camps.

Measurable outcome:

The system can register the community member pre-disaster and verify their identities in real-time as they gain entry to the camps with a fingerprint accuracy rate of 95% or above on a test data set under hardware constraints

Aligned component: Fingerprint Registration Module Wristband Issuance Module

Objective 2

To create a Source Action Classification Schema (SACS) that automatically assigns levels of criticality Critical, Important, Routine to all system-generated MQTT messages based on the type of source from which the message was initiated and the action that initiated the message in the Smart Camp Box architecture.

Measurable outcome:

Schema correctly classifies 100% of defined message source-action combinations across all six system source types: fingerprint scanner, wristband scanner, web form, admin dashboard, prediction model, and system internal with zero misclassification on defined rules.

Objective 3

To design and implement a real-time network state monitoring mechanism that continuously measures network latency and packet loss rate, classifying connectivity into four operational states Good, Weak, Poor, and Offline to inform dynamic communication decisions.

To develop a mechanism for the implementation of a real-time network state monitoring system, which can measure network latency as well as network packet loss rate, thereby classifying the network state into four categories, namely, Good, Weak, Poor, and Offline, for the purpose of making dynamic communication decisions.

Measurable Outcome: Network monitor successfully identifies simulated network states with a classification accuracy of 90% or higher when tested under controlled network degradation environments, including 0%, 15%, 30%, and 100% network packet loss

Aligned Component: Message Router Network Monitor Module (Novelty Point 2)

Objective 4

To design and test the Semantic-Aware Adaptive Message Prioritization framework for the dynamic allocation of MQTT Quality of Service levels and message transmission order based on the combined results of the Source Action Classification Schema and the real-time network state monitor, with offline buffering and cloud synchronization in a prioritized order when a network connection is re-established.

Measurable outcome:

Critical message delivery rate of 95% or more under simulated weak and poor network conditions, clearly surpassing static MQTT QoS 0, QoS 1, and QoS 2 baselines in terms of:

Critical message delivery rate (%)

Message latency (ms)

Packet loss rate (%)

Priority corrects flush order (%)

when a network connection is re-established

Aligned component:

Message Router Full Framework (Core Novelty Contribution)

Objective 6

To create an offline-enabled subsystem for delivering displaced people in camps awareness video content and documents via local HTTP connectivity and real-time administrative announcements via a WebSocket-enabled MQTT bridge, independent of internet connectivity.

Measurable Outcome:

The subsystem should successfully serve video content to at least 20 connected devices at a time via local Wi-Fi network without internet connectivity and should serve WebSocket announcements to all connected clients in less than 2 seconds from publication via MQTT.

Aligned Component: Awareness and Education Module WebSocket Bridge

3. METHODOLOGY

3.1 Research Approach

The research approach that is adopted for this study is that of Design Science Research (DSR), which is arguably the most accepted and suitable framework for information technology and computing-related research that involves developing a novel technology artifact as its primary contribution. The Design Science Research methodology for information systems is based on a six-step process that starts with problem identification and motivation, followed by specifying the objectives of a solution, designing and development, demonstration, evaluation, and communication, which are directly applicable to developing and evaluating IoT-based systems. DSR is chosen over other approaches like experimental or survey-based methods for this study due to three specific reasons related to this research: first, this study is primarily focused on developing a technology artifact, namely, the prototype of the Smart Camp Box system along with its Semantic-Aware Adaptive Message Prioritization framework, instead of a theoretical model or statistical analysis. Second, this study is focused on a real-world operational scenario involving disaster relief camps in Sri Lanka that needs a demonstrable solution instead of just an analytic contribution. Third, DSR supports iterative development and prototype-based evaluation. This is important considering the 7-month schedule and the integration of hardware and software components. The reason for this is that prototyping is a good fit for IoT system development. The reason for this is that end-users and stakeholders do not really have a good understanding of the requirements for an IoT system until they use the product itself. The integration of hardware and software into an IoT system is complex.

PHASE 1 DESIGN (Months 1–2)

- Confirm literature gap
- Finalize system architecture
- Finalize algorithm specification (SACS + Adaptive QoS)
- Finalize hardware and software selection

PHASE 2 DEVELOPMENT (Months 3–4)

- Implement each component
- Integrate all the modules
- Unit test each component
- Perform end-to-end integration testing

PHASE 3 EVALUATION (Months 5–6)

- Design a controlled experiment
- Perform baseline comparison experiments
- Measure the performance
- Perform analysis, interpretation, and thesis writing.

PHASE 4 EVALUATION

- Controlled experiment design
- Baseline comparison testing
- Performance measurement
- Output: Quantitative results

3.2 System Architecture Design

3.2.1 Overall Architecture Philosophy

The Smart Camp Box system is based on a philosophy that prioritizes edge computing and being able to operate offline or in a disconnected mode. This means that internet connectivity is viewed as a nice-to-have rather than a must-have. This architectural philosophy is particularly relevant for the infrastructure challenges faced during a disaster event in Sri Lanka, where internet connectivity is unreliable, bandwidth is severely constrained, and reliance on cloud computing is viewed as a significant operational risk. The Smart Camp Box architecture is based on a three-tier model:

TIER 1 LOCAL EDGE LAYER

- All primary operations are executed locally.
- No internet connectivity requirements.
- The IoT device functions as a fully standalone system.
- Local Wi-Fi network for camp residents.

TIER 2 COMMUNICATION LAYER

- MQTT broker for message routing internally.
- Adaptive QoS for transmission.
- WebSocket bridge for connected clients.
- HTTP server for serving static content locally.

TIER 3 CLOUD SYNCHRONISATION LAYER

- Opportunistic sync on connectivity.
- Priority-ordered message flushing on reconnects.
- Cloud storage for historical data and analytics.
- Machine learning, model training, and inference.

3.2.2 Hardware Platform

The main edge device is a Raspberry Pi 4 Model B, 4GB RAM variant, chosen for its balance of processing performance, power consumption, interface capabilities, and prior use in IoT-related research deployments. The device runs all local system components, including the MQTT broker, backend server, database, and network monitoring module. A WiFi access point module is integrated into the system for creating the local network that camp residents' personal devices connect to.

The peripheral devices are a USB fingerprint reader module that supports ISO/IEC 19794-2 compliant template storage, and a USB barcode reader for reading barcodes on resource distribution point wristbands. The symbology of barcodes on wristbands is based on the Code 128 standard, chosen for its high data density and robustness against wear during camp usage.

3.3 Technical Methods and Algorithms

3.3.1 Biometric Registration and Verification Method The proposed system utilizes a two-phase model for its operation. In the pre-disaster phase, fingerprint template creation, processing, and

storage are performed locally on the edge device through a one-to-many matching approach. In the disaster time phase, one-to-many matching is performed against the locally stored template database to identify registered individuals returning to the site, and creation of new templates for individuals not registered prior to the disaster. The storage of fingerprint templates utilizes a non-reversible transformation approach for securing the data during storage. The raw data of the minutiae template is not directly stored, but instead a non-reversible transformation of this data is performed, which cannot be used to recreate the original template, following the principles of cancelable biometrics proposed in the literature. The issuance of the wristband takes place immediately after successful verification or creation of a new template. The ID of the wristband is securely related to the person profile in the local database, creating a chain of custody between the biometric identity and the physical token utilized for resource distribution interactions.

3.3.2 Source-Action Classification Schema (SACS)

The Source-Action Classification Schema is the first element of the central novelty framework. The Source-Action Classification Schema is a formally defined rule-based system for message classification that assigns a criticality level to all messages generated within the Smart Camp Box system based on two attributes: the source type of the message and the specific action that triggered message generation.
The schema consists of six source types corresponding to the six message-generating components of the system architecture. For each source type, a finite set of possible message-generating actions is defined. A criticality level is assigned to each source-action pair based on the outcome of message loss in the context of the disaster relief camp. The theoretical foundation for SACS is based on the assignment of criticality levels based on three criteria for each source-action pair:

Criterion 1 - Recoverability: The ability to recover or reconstruct the data contained within the message in the event of message loss. The data contained within the biometric registration records and message confirmations is not recoverable based on system data alone; loss of such messages results in permanent loss of information.
Criterion 2 Consequence Severity: The impact that the loss of the message has upon the well-being of displaced individuals. Those whose loss could mean that an individual would not receive food, water, or medical attention are at the highest level of criticality.
Criterion 3 Temporal Irreplaceability: Whether the message represents a unique occurrence that cannot be regenerated by the system. Heartbeat messages are regenerated continuously, and their loss is inconsequential; biometric verification is a one-time event.

3.3.3 Network State Monitoring Method

The network state monitoring component employs a continuous multi-metric measurement method that measures and evaluates the network performance metrics at regular intervals to identify the network connectivity state. The network performance metrics measured by the network state monitoring component include:

Metric 1 Round-Trip Latency: The network performance metric is measured by periodically sending a probe request to a known endpoint in the cloud. The network performance metric is measured in milliseconds. The network performance metric is a good measure of network responsiveness and is the least computationally expensive to measure.

Metric 2 Packet Loss Rate: The network performance metric is measured by transmitting a predetermined number of packets over a given time. The network performance metric is a good measure of network reliability and is independent of network responsiveness.

Metric 3 MQTT Broker Acknowledgement Time: This is defined as the time taken to transmit a QoS 1 message to the local Mosquitto MQTT broker, measured as the time elapsed before receiving the PUBACK message for the corresponding message. The time is measured in milliseconds.

The three metrics are used to determine one of the following discrete network states: Good, Weak, Poor, or Offline. The thresholds used for the metrics are based on a weighted decision function. The thresholds are derived based on empirical data during the system test phase of the framework. The thresholds are documented as system parameters to adapt to different environments.

3.3.4 Semantic-Aware Adaptive QoS Decision Framework

The Semantic-Aware Adaptive QoS Decision Framework represents the technical innovation of this study. The Semantic-Aware Adaptive QoS Decision Framework is a message routing middleware layer that is placed between the system components that generate the message and the Mosquitto MQTT broker. The middleware layer intercepts all outgoing messages and applies a two-dimensional decision process before message transmission. The decision process has two inputs: the criticality level defined by the SACS and the network state classification obtained from the network monitor. The decision process applies to a decision matrix that has all combinations of the criticality level and network state classifications. The decision matrix is formally defined and has an output that consists of three elements: the QoS level that will be assigned to the message, the disposition of the message for transmission, and the priority for message buffering.

The framework also regulates the offline buffering mechanism. In the case where the network is classified as Offline, all messages to be sent are written to a local priority queue data structure stored persistently on the storage device of the edge device. The criticality level of each message is preserved as a key for the priority queue. Upon the detection of a restored network connectivity condition via a successful network state recategorization to any network state other than Offline, a controlled flush of the messages is performed in a manner that processes messages in descending order of their criticality levels, where all Critical messages are fully acknowledged before any Important message transmission begins, and all Important messages are fully acknowledged before any Routine message transmission begins.

The process ensures that the most critical data is transmitted to the cloud first when bandwidth is still a limiting factor even after restored connectivity is achieved, which is a realistic assumption for the recovery process of Sri Lanka’s disaster events.

3.3.5 Resource Prediction Method

The cloud-based resource prediction module utilizes supervised learning to analyze historical registration and distribution data to forecast daily resource requirements for camp administrators. The forecasting targets are three continuous data values: daily food unit requirements, daily water volume requirements, and daily medicine kits.

The features are based on four categories of input data: demographic composition of the registered population within the camp, including age group, gender group, and medical conditions; historical patterns of resource consumption based on previous distribution cycles; temporal patterns including duration of disasters and seasonal context; and population patterns including daily population intake rates and population departure rates.

The selection of the best model is based on a comparative evaluation approach to train multiple regression-based models on a validation data set created by combining publicly available disaster relief distribution data sets with synthetic data created using UNHCR operational statistics. The best model is the one with the minimum Mean Absolute Percentage Error on the validation data set.

3.4 Tools and Platforms with Justification

3.4.1 Communication Layer

Mosquitto MQTT Broker is used as the local message broker due to its position as the most widely used open-source MQTT broker, its proven reliability on the Raspberry Pi platform, its native support for persistent sessions and quality of service levels 0 to 2, and its bridge capabilities that can be used for direct integration with the cloud EMQX MQTT broker. The low memory footprint of Mosquitto is essential for edge device deployment. EMQX Cloud MQTT Broker is used as the cloud layer for the reception of messages due to its reliability, its ability to handle a high volume of messages, its native compatibility with MQTT bridges for Mosquitto, and its built-in monitoring dashboard. The EMQX Cloud MQTT Broker is used on a virtual machine for a controlled cloud endpoint for experiments. ***WebSocket Protocol*** is used for the delivery of real-time announcements to the devices of the residents of the camp due to its full-duplex communication, its compatibility with a wide range of mobile browser environments without the need for any application installation, and its ability to maintain a persistent connection over the local Wi-Fi network regardless of internet connectivity.

3.4.2 Backend and Frontend

Node.js/Express.js is chosen for backend server development due to its event-driven, non-blocking I/O model that is suitable for handling multiple MQTT message events, HTTP content requests, WebSocket connections, and barcode scanner input events concurrently, which is in line with the nature of Smart Camp Box system operation. Node.js also supports native MQTT client libraries through its mqtt.js package, allowing close integration between HTTP server functionality and message routing functionality.

React.js is chosen for building the frontend web interface due to its ability to support a component-based development model that allows for separate development of the resource request form, admin dashboard, and educational content portal as separate entities. React.js also supports mobile browsers, allowing camp residents to access the system interface through consumer smartphones without requiring any application installations.

SQLite is chosen for local database functionality due to its serverless architecture, zero-configuration requirements, and its ability to support single-device edge computing scenarios. SQLite provides adequate transactional integrity for local storage of biometric profiles, entitlements, and distribution records, yet imposes minimal resource requirements on the edge of device hardware.

3.4.3 Cloud and Machine Learning

Python and scikit-learn were chosen for the development of the resource prediction model due to the extensive library available for machine learning algorithms and the simplicity of deployment as a RESTful API service. Python is also chosen for data processing and feature engineering for the historical data using pandas library. Firebase Firestore or similar cloud-based NoSQL data storage solutions were chosen for the storage of synchronized records in the cloud environment based on their real-time synchronization capabilities and scalability in storing different document formats for the records obtained from the Smart Camp Box system.

3.4.4 Development and Testing

The process of network emulation during evaluation experiments is carried out using tc (Traffic Control) on Linux, which allows precise control over network latency, packet loss rate, and bandwidth limitation during evaluation experiment processes, without requiring physically deteriorated network infrastructure for disaster scenarios.

Wireshark is used for capturing and analyzing network packets during evaluation experiment processes, allowing actual verification of MQTT QoS behaviour, packet loss, and ordering of messages during experiment processes. Postman is used for testing REST API and validating cloud synchronisation endpoints during system integration testing processes.

3.5 Data Collection and Ethical Considerations

3.5.1 Data Collection Strategy

The data collection activity involved in the proposed research happens on two distinct levels: system performance data collected during the evaluation experiments, and registration and distribution data used to train the prediction models.

System Performance Data is collected via the logging instrumentation provided as a component of the Smart Camp Box. Every message transmission event, every decision to assign a quality of service to a message, every network state identification event, every buffering event, and every queue flush event are all recorded to a log file along with a high-resolution timestamp. This provides a complete audit trail of the entire system under each set of experimental conditions.

Prediction Model Training Data is collected under three types of publicly available UNHCR operational data on the demographics of the population residing in refugee camps and resource consumption statistics, publicly available Sri Lanka Disaster Management Centre situation reports on the demographics of the population residing in disaster relief camps and distribution statistics on past disaster events, and synthetic data derived via statistical simulation using the parameters of past disaster events. There is no collection of personal data on actual individuals.

Usability and Functional Testing Data is collected through a series of structured functional testing sessions where participants, who are university students, volunteer their time to perform a simulation of the camp registration, verification, and resource request process. Participants are fully aware of the purpose of the research prior to taking part in the study.

3.5.2 Ethical Considerations

The study will involve the collection of biometric data during the process of conducting functional tests. This implies the need for certain ethical considerations to be considered while conducting the study, considering the Sri Lankan regulations regarding personal data protection.

Informed Consent: The study will ensure that all individuals who take part in the process of conducting functional tests have given their informed consent before any data collection process is initiated.

Protection of Biometric Data: The fingerprint data collected during the process of conducting functional tests will be stored only in the transformed but irreversible format on the device used for conducting the study. The raw images of the biometric data are not stored.

Data Minimization: The study only collects the minimum amount of biometric data required to test the functionality of the system. Research Ethics Approval: The study is approved by the university's institutional research ethics committee. The data collection process is done while considering the terms and conditions of the approval.

3.6 Validation Strategy and Performance Metrics

3.6.1 Validation Strategy

Validation of the Smart Camp Box framework is conducted through a **controlled comparative experimental design** in which the proposed Semantic-Aware Adaptive Message Prioritization framework is evaluated against three baseline configurations under identical controlled network degradation conditions. This comparative approach is consistent with established practice in IoT communication protocol evaluation and provides the quantitative evidence necessary to demonstrate that the proposed framework produces measurably superior outcomes relative to existing approaches.

The three baseline configurations against which the framework is compared are:

Baseline 1 Static QoS 0: All messages transmitted with MQTT QoS level 0 regardless of content or network conditions, representing the minimum viable MQTT implementation and the most deployed configuration in resource-constrained IoT systems.

Baseline 2 Static QoS 2: All messages transmitted with MQTT QoS level 2 regardless of content or network conditions, representing the maximum reliability standard MQTT configuration and the approach implicitly recommended for critical data by the MQTT specification.

Baseline 3 Static Topic-Based Priority: Messages assigned QoS levels based on manually predefined topic labels without network state awareness, representing the current best practice in priority-aware MQTT deployment as described in the existing literature.

Each scenario is executed for a specified time with a standardised message workload that simulates the realistic operation of the Smart Camp Box. The message workload is defined as a percentage of Critical, Important, and Routine messages with realistic inter-arrival times. The experiments are performed five times for each configuration for each scenario to account for measurement variability. The results are reported as mean values with standard deviations.

3.6.2 Performance Metrics

The evaluation will consider six performance metrics that jointly describe the performance of each configuration across the experimental scenarios:

Metric 1: Critical Message Delivery Rate (CMDR)

The percentage of Critical messages that are acknowledged by the cloud broker within the experiment duration. This is the first metric to be evaluated as it is the primary evaluation metric for the framework's core objective: protecting critical data under network degradation. A higher value for the CMDR is desirable.

Metric 2 - End-to-End Message Latency (EEL)

The total time taken in milliseconds between the generation of a message from the source component and the receipt of the acknowledgement from the cloud broker for each criticality level. A lower latency for Critical messages compared to Routine messages is a clear indication that priority has been enforced correctly.

Metric 3 - Overall, Packet Loss Rate (PLR)

The percentage of total messages sent that fail to receive a cloud acknowledgement within the experiment duration for each criticality level. This measures whether the percentage of lost packets is evenly distributed across criticality levels or predominantly for Routine messages.

Metric 4 - Queue Flush Priority Compliance Rate (QFPCR)

The percentage of messages delivered by the component during the intermittent connectivity experiment that are delivered strictly in descending order of criticality levels. This measures the effectiveness of the buffering mechanism for offline messages.

Metric 5 - Network State Classification Accuracy (NSCA)

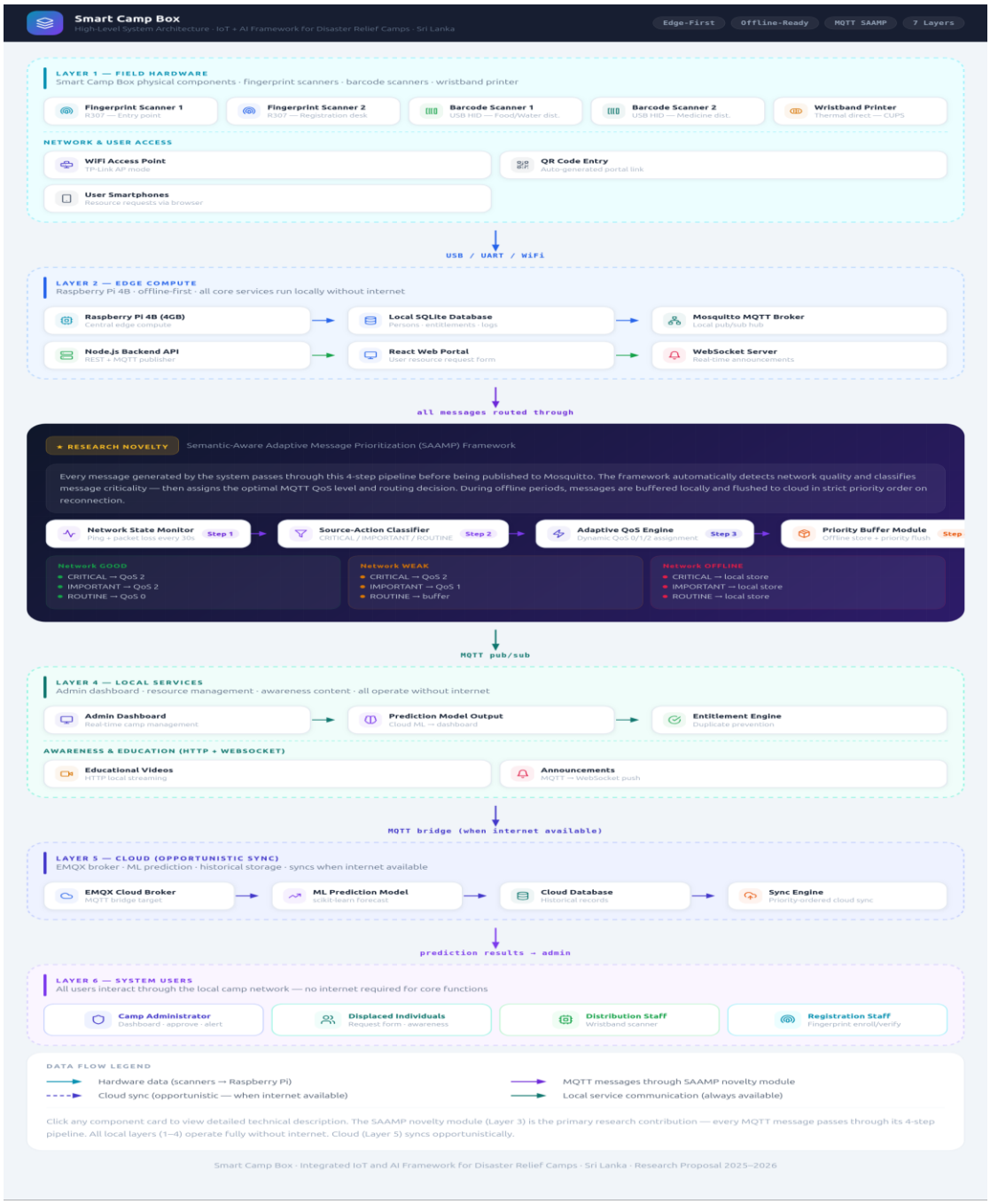
The percentage of network states classified by the monitoring module that match the simulated network states provided by the ground truth network state provided by the traffic control emulation tool.

3.6.3 Success Criteria

The proposed framework is considered to have demonstrated its research contribution if the following minimum success criteria are satisfied across the evaluation experiments:

Metric	Minimum Success Criterion
Critical Message Delivery Rate	≥ 95% under Scenarios 2, 3, and 4. Must outperform all three baseline methods.
End-to-End Latency (Critical)	Must be lower than **Static QoS 0** and **Static Topic-Based Priority** under Scenarios 2 and 3.
Packet Loss Rate (Critical)	≤ 5% under Scenarios 2 and 3. Must be lower than all three baselines.
Queue Flush Priority Compliance	≥ 95% under Scenario 4
Network State Classification Accuracy	≥ 90% accuracy across all scenarios.
Bandwidth Consumption Rate	Must be lower than **Static QoS 2 baseline** under Scenarios 2, 3, and 4.

4. HIGH-LEVEL SYSTEM ARCHITECTURE



5. USER REQUIREMENTS

5.1 Stakeholder identification

For an effective disaster relief system using the IoT technology, proper planning in a country-specific context involving different stakeholders is essential, as visualisation of the actual disaster scenario by technical people alone is not enough to make a technical solution effective at the ground level. The Smart Camp Box system caters to a set of primary and secondary stakeholders who have a specific need and interaction with the system that ultimately contributes to the requirements discussed in this section.

2.2 Functional Requirements

Functional requirements define what the system must do. All requirements are numbered for traceability and mapped to the stakeholder they serve.

Module 1 Pre-Disaster Fingerprint Registration

Functional Requirement ID	Functional requirement
FR-01	The system should enable registration officers to register community members before disaster strikes through a fingerprint scanner connected to the IoT edge device.
FR-02	The system should collect and store an individual's profile information such as name, age, gender, National Identity Card number, medical conditions/special needs, number of family members, and home location/administrative zone before disaster strikes.
FR-03	The system should store the fingerprint templates and associated user data locally in the IoT edge device's database as soon as an individual is enrolled without the need for an internet connection.
FR-04	The system should synchronize the locally stored fingerprint registration data with the cloud database as soon as an internet connection is available.
FR-05	The system should check for duplicate registration by matching the newly captured fingerprint data against existing fingerprint templates before a new user profile is created.

Why it is a user requirement:

Pre-disaster fingerprint registration is recognized as a user requirement due to the identification of the requirement from the operational requirement of the registration officers. The government of Sri Lanka has a mandate to prepare for disaster situations. Effective disaster management is a requirement that is best addressed with adaptive IoT architectures that incorporate socio-technical integration with various stakeholders such as the government. The fingerprint enrollment process is the core user requirement on which all other functions are based.

Module 2 Disaster-Time Identity Verification and Registration

FR-06	The system should verify individuals arriving at the disaster relief camp by matching the fingerprint against the pre-registered fingerprint template database stored in the local IoT edge device.
FR-07	Upon successful verification of the fingerprints, the system should display the complete individual profile within 3 seconds from the time the fingerprint scanning is completed.
FR-08	In case the fingerprints do not match the existing templates in the database, the system should allow individuals to register by creating a new record and collecting the individual's basic profile information via the web interface for camp entry.
FR-09	Upon successful registration of a new individual, the system should publish an MQTT message to the topic camp/persons/registered with a priority of CRITICAL to notify other system components.
FR-10	Upon successful verification of a pre-registered individual's identity, the system should publish an MQTT message to the topic camp/persons/verified with a priority of CRITICAL to notify other system components.

Module 3 — Wristband Issuance and Management

FR-11	After successful identity verification or registration, the system shall generate a unique wristband identifier and map it to the corresponding individual profile.
FR-12	The system shall store entitlement information associated with the wristband profile, including daily food allocation (meals per day), daily water allocation (liters per day), medicine entitlement (availability and type), etc., and special needs accommodations.
FR-13	The system shall provide the facility to re-issue the wristband by re-verifying the individual's identity through fingerprint authentication when the wristband is found to be lost or damaged. On re-issue of the wristband, the original wristband identifier shall be invalidated and replaced by a newly generated one associated with the same individual profile.
FR-14	On re-issue of the wristband, the system shall send an MQTT message to the topic camp/persons/wristband/replaced under CRITICAL priority.

FR-15	The system shall classify every MQTT message generated by any system component into one of three priority levels: CRITICAL, IMPORTANT, or ROUTINE, based on the combination of message source type and triggering action using the Source-Action Classification Schema (SACS).
FR-16	The system shall continuously monitor the real-time network state by measuring round-trip latency (in ms) every 30 seconds, packet loss rate (%) using a sliding window, and MQTT broker acknowledgement time (in ms).
FR-17	The system shall classify the current network state into one of four states based on the measurement of network metrics: GOOD (latency < 100 ms, packet loss < 5%), WEAK (latency between 100-500 ms, packet loss between 5-20%), POOR (latency > 500 ms, packet loss > 20%), or OFFLINE (no response, MQTT broker not reachable).
FR-18	The system shall dynamically classify MQTT Quality of Service (QoS) levels of outgoing messages based on the combined evaluation of message priority classification and current network state based on the decision matrix provided in Table X.
FR-19	The system shall ensure that CRITICAL messages are never dropped under any network condition and shall be kept locally until successful delivery is confirmed to the cloud broker.
FR-20	The system shall ensure that the messages labeled as CRITICAL are never lost under any network condition, and they shall be kept locally in the storage until the successful delivery to the cloud broker is confirmed.

Non-Functional Requirements

Non-functional requirements define how the system must perform, covering security, performance, usability, reliability, and scalability.

Performance Requirements

NFR-01– Fingerprint matching on edge device hardware must be completed in 3 seconds or less during normal load conditions.

NFR-02 Wristband barcode scanning for distribution authorization must be completed in 2 seconds or less during normal operation.

NFR-03 MQTT message delivery latency for CRITICAL messages must be 500 milliseconds or less during GOOD network conditions.

NFR-04 The web interface must load in 3 seconds or less on any smartphone that is connected to the local camp Wi-Fi network.

NFR-05 The system must support a minimum of 200 concurrent Wi-Fi connections from user devices without performance degradation.

Usability & Accessibility Requirements

NFR-01 – The user interface for the web application must be usable by individuals without any prior technical knowledge in less than 5 minutes of first usage.

NFR-02 – The user interface must support three different languages for users from the three major linguistic groups in Sri Lanka: Sinhala, Tamil, and English.

NFR-03 – The user interface for the distribution terminal must include visual and audible feedback for the results of the scanning action:

- Green for authorized scan results
- Red for denied scan results
- Alerting sound for duplicate results

NFR-04 – Educational material must be accessible from any smartphone browser without the need for application installation or user account creation.

Usability & Accessibility

NFR-01 -Usability
The system shall offer a web interface that can be effectively used by individuals who do not have technical knowledge. The system should be effective 5 minutes from the start of interaction.

NFR-02 -Language Support
The system shall support Sinhala, Tamil, and English languages to cover all linguistic groups in Sri Lanka.

NFR-03 - Feedback for Distribution Terminal
The system shall offer visual and audible feedback for the distribution terminal for the results of the scans as follows:

- Green for authorized scans

- Red for denied scans
- Alerting sound for duplicate detection

NFR-04 - Educational Content Accessibility
The system shall offer access to educational content via any smartphone browser without the need for installing an application or registration.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona The primary target market for the Smart Camp Box system consists of government organizations in Sri Lanka that are responsible for managing disaster relief operations. The key customers and their characteristics are presented in the table below:

Institution 1 National Disaster Relief Services Centre (NDRSC) - Ministry of Disaster Management

- **Decision Maker** Director General - NDRSC
- **Budget Source** Annual government allocation
- **Pain Points** Manual camp registrations; manual records for distribution; lack of a digital coordination platform
- **Buying Triggers** Failure in audit reports post-disaster; media reports criticizing gaps in relief operations.
- **Procurement Method** Tender from the government

Institution 2 Disaster Management Centre (DMC) - Ministry of Public Administration

- **Decision Maker** Director General - DMC
- **Budget Source** Government allocation; additional funding from international sources
- **Pain Points** No digital platform for camp population data; insufficient fraud controls for resource allocation
- **Buying Triggers** Donor reporting requirements; need for donor accountability

Institution 3 District Secretariats - all 25 districts in Sri Lanka

- **Decision Maker** District Secretary or Divisional Secretary
- **Budget Source** Allocation from provincial council
- **Pain Points** Manual camp management during monsoon season
- **Buying Triggers** Annual monsoon season; need for proactive disaster management

Value Proposition

Target Users: Government agencies for disaster relief, humanitarian organizations, NGOs, etc., in Sri Lanka and other disaster-affected nations.

User Need: To provide reliable, fraud-free human identity management services along with tracking resource distribution in disaster relief camps without requiring internet connectivity.

The Smart Camp Box Solution: A self-sufficient IoT solution that
creates its own network for offline operation and provides biometric-based human identity management services for all camp dwellers.
Guarantees that no critical data is lost under any circumstances, including internet connectivity failure.
Differentiation: The solution is differentiated from existing paper-based methods that do not offer any audit trail or cloud-based methods that fail when internet connectivity is not present.

Key Advantages:

- Guarantees zero critical data loss under any condition.
- Provides fraud-free resource distribution through wristband and fingerprint-based verification.
- Allows offline operation indefinitely.
- Enables instant deployment within 30 minutes of reaching a disaster site.
- Allows proactive resource management through predictive analysis of future needs.

Revenue Model

The commercialization model for the Smart Camp Box is based on a three-stream revenue model. The revenue model is suitable for humanitarian organizations and government procurement. The model is explained in detail below:

Revenue Stream 1: Hardware Sales (Unit Sales Model)

Product: Smart Camp Box physical unit consisting of:

- Raspberry Pi 5 edge compute
- Fingerprint scanner module
- Barcode scanner
- Wi-Fi access point hardware
- Local storage: SSD
- Power bank/solar input port
- Ruggedized weatherproof casing

Target Customers: Government institutions such as NDRSC and DMC. Humanitarian organizations such as UNHCR and WFP.

Business Model: One-time purchase per unit. Bulk discounts for 10+ units. Optional maintenance contract.

Revenue Stream 2: Software Licensing (Annual SaaS)

Product: Cloud platform subscription consisting of:

- EMQX cloud broker hosting
- Predictive analytics for resource management
- Historical data storage
- Multi-camp operational dashboard
- Software updates and patches

Target Customers: All hardware customers

Business Model: Annual license fee per active unit. Tiered pricing for organization size.

7. BUDGET AND JUSTIFICATION

Hardware Costs

All hardware listed below forms the physical Smart Camp Box prototype required for development, testing, and evaluation of the proposed research.

Item 1 Raspberry Pi 5 (4GB) Edge Computing Unit

Purpose:
The Raspberry Pi 5 serves as the core processing unit of the Smart Camp Box, running all software components, including:

- Mosquitto MQTT broker
- Node.js backend
- React web server
- Fingerprint matching engine
- Network monitoring module
- Novelty message routing module

Quantity: 1 unit

Unit Price: LKR 27,000

Justification for Selection:

The Raspberry Pi 5 is selected over other lower-cost options (e.g., Raspberry Pi 4, Arduino) for the following reasons:

Processing Power: It has a powerful quad-core 2.4 GHz processor that does not compromise performance even if all operations like fingerprint matching, MQTT broker, and web server are run simultaneously.

Memory: It has 4 GB of RAM that is sufficient for all the components to run simultaneously.

Connectivity: It has built-in Wi-Fi that supports access point mode; hence, a local camp network can be set up on the Raspberry Pi.

Peripheral Support: It has USB 3.0 ports that allow for simultaneous connection of the fingerprint scanner and the barcode scanner.

Item 2 Fingerprint Scanner Modules (R307/R503)

Purpose: Fingerprint scanner modules are used for fingerprint template matching for pre-disaster registration and disaster-time identity verification purposes. These fingerprint scanner modules are the main hardware components for supporting Functional Requirements FR-01 to FR-10.

Quantity: 2 units

Unit 1: To be used as an entry point for disaster-time verification of individuals entering the camp.

Unit 2: To be used as a registration desk for new on-site registrations during disaster events.

Unit Price: Estimated LKR 3,500 (R307/R503 grade).

Source: TRONIC.LK Sri Lanka – R307 and R503 fingerprint scanner modules are available in Sri Lanka.

Total Cost: LKR 7,000

Justification for Quantity: We require two units of fingerprint scanner modules for smooth operations:

Unit 1:

- To be used as an entry point for disaster-time verification of individuals entering the camp.

Unit 2:

- To be used as a registration desk for new on-site registrations during disaster events.
- If we use a single unit, it will cause a bottleneck for individuals in the queue, compromising efficiency during mass influx events.

Specification Requirements:

- Minimum capacity of 1,000 fingerprint templates (R307 fingerprint scanner module: 1,000 capacities)
- R503 fingerprint scanner module: 200 capacities.
- TTL UART interface – compatible with Raspberry Pi GPIO.

Item 3 - USB Barcode Scanner (1D/2D Handheld)

Purpose:

The USB barcode scanner is employed at the resource distribution stations in the camp, where the barcodes on the wristbands are scanned. Each scan operation initiates the logic for checking whether the scanned individual is valid, whether he/she has consumed the resource on that day, and whether he/she is eligible for that resource. Scan operations generate the highest priority MQTT messages, namely, 'CRITICAL' priority messages, for the 'resource distribution' subsystem, specifically for topics 'camp/resources/distributed' and 'camp/resources/denied'.

Quantity:

2 Units

Unit 1:

The unit is employed at the primary station for the distribution of resources, i.e., the food and water distribution station.

Unit 2:

- The unit is employed at the medical and shelter distribution station.
- Unit Price:
- Estimated in LKR 4,500

Source: Elecshop.lk Sri Lanka as the USB barcode scanners are readily available in the market in this price range.

Total Cost: LKR 9,000

Justification for Quantity:

There are two distribution points in a functional relief camp that operate simultaneously. These are the food and water distribution points, and the medical and shelter distribution points. A single scanner for both distribution points would mean that the staff would have to move from one point to the other with each scan event, thereby causing an unreasonable delay in the distribution of resources. Having two scanners ensures the uninterrupted operation of both distribution points.

Specification Requirements:

USB HID Interface:

The USB HID interface of the scanner should be plug-and-play compatible with the Raspberry Pi USB port, with no need for driver installation.

Minimum 1D barcode Support: The scanner should be able to scan a minimum of 1D barcode fonts such as Code 128, Code 39, EAN-13, which is the format of the wristband barcode.

Scan Rate:

- The scanner should have a scan rate of a minimum of 100 scans per second.
- Linux OS Support:The scanner should be compatible with the Linux operating system.

Item 4 - Wristband Barcode Labels (Pre-printed Roll)

Purpose: Wristband barcode labels are the physical identity tokens provided to individuals entering the disaster relief camp after fingerprint authentication or new registration. Each of these labels carries a unique barcode linked to the individual's database profile. This wristband acts as the primary interface for interactions related to the distribution of resources to the individuals in the disaster relief camp, eliminating the need for fingerprint authentication at multiple points of resource distribution.

Quantity: 500 Units (1 roll - sufficient for the purpose of the prototype simulation and testing)

Unit Price: Estimated Cost - 0.60 LKR (Thermal Direct Wristband Label Stock - Standard 25mm x 280mm size roll stock)

Source: Barcode Lanka Holdings (barcodelanka.lk) Sri Lanka / DB Automation (Pvt) Ltd Sri Lanka - Wristband Label Rolls are readily available.

Total Cost: 300 LKR (for 500 Units)

Justification for the Quantity:

The quantity of 500 Units of Wristband Barcode Labels would be sufficient for the purpose of the prototype evaluation phase of the proposed research, as it would enable the simulation of the population tests, stress tests, and the experiments to be conducted for the purpose of validating the Semantic-Aware Adaptive Message Prioritization framework.

Specification Requirements:

- Direct thermal print format (no ink required, compatible with standard thermal label printers)
- Minimum 25mm width this will be enough for barcode readability at the scanning distance expected for a distribution scenario.
- Water-resistant material will be a necessity for the outdoor disaster camp scenario.
- Adhesive closure or snap fit band format, adult and child wrist sizes.

Item 5 – Portable Power Bank (High Capacity)

Purpose:

The purpose of the portable power bank is to ensure an uninterrupted power supply to the Raspberry Pi 4, Wi-Fi router, and fingerprint scanners in disaster scenarios where electricity is not available or is interrupted. The need for an uninterrupted power supply is an essential requirement for the offline-first edge computing architecture, and any interruption in the power supply to the camp device equates to an outage of the complete system, irrespective of the optimizations carried out in the communication protocol.

Quantity:

1 Unit

Unit Price: Estimated LKR 8,000 (30,000mAh dual-output power bank, enough for 8-12 hours of continuous operation)

Source: Daraz.lk Sri Lanka, local electronics retailers

Total Cost: LKR 8,000

Justification for Quantity:

The requirement for the power bank is based on the assumption that one high-capacity power bank, 30,000mAh, will be enough to power the complete Smart Camp Box system for 8-12 hours, thus equating to one complete day's operation between charge cycles, matching the operational rate of disaster relief camp distribution scenarios.

CLOUD SERVICES

Item 6 -Cloud Virtual Machine (EMQX Cloud Broker + Prediction Model Hosting)

Purpose: The cloud virtual machine will be used to host two services. First, the EMQX MQTT broker will receive bridged messages from the local Mosquitto broker when internet connectivity is present. Second, the prediction model will be hosted to analyze the registration and distribution data to predict the future requirements for the admin dashboard. The cloud layer will be the target for the offline-first edge architecture, and it will be the storage repository for all the camp operational data.

Quantity: 1 VM Instance × 7 months (research duration)

Specification: 2 vCPU/4 GB RAM/50 GB SSD (similar to AWS t3.medium or Google Cloud e2-medium) - enough for EMQX broker and ML model

Unit Price: Approximately 15-20 USD/month (4,800-6,400 LKR/month based on current exchange rate)

Source: Oracle Cloud Infrastructure (has free tier with 1 OCPU/1 GB RAM VM - can be used for prototype phase free of charge)

Total Cost: LKR 0 (Prototype Phase - Using Oracle Cloud Free Tier) - LKR 44,800 (maximum possible cost for 7 months)

HUMAN EFFORT COST

Item 7 Research and Development Effort

Purpose: This item captures the estimated human effort invested in the research, system design, software development, hardware integration, evaluation, and academic writing components of the project across the 7-month research timeline.

8. WORK BREAKDOWN STRUCTURE (WBS)

Task Breakdown Per Member

Member 1 – Hansana G P | IT22064868

Sub-Component: Smart Environmental Risk Monitoring and Location-Aware Flood Prediction

- **SO1 – Automated Geospatial Data Acquisition**
  Implement a high-precision GPS/GNSS module to fetch and transmit real-time latitude and longitude coordinates for each disaster camp.
- **SO2 – Digital Elevation Model (DEM) Integration**
  Integrate a DEM to extract altitude and topographical vulnerability data based on camp coordinates.
- **SO3 – Real-time Environmental Data Telemetry**
  Develop an IoT sensor node to monitor rainfall intensity, soil moisture, and humidity continuously.
- **SO4 – Algorithmic Gauging Station Identification**
  Engineer a proximity-based algorithm to identify the nearest Irrigation Department gauging station.
- **SO5 – External Data Scraping & API Integration**
  Develop a web scraper/API integration to retrieve river level data and weather forecasts.
- **SO6 – Geospatial Watershed & Hydrological Analysis**
  Perform watershed and runoff analysis using QGIS and OpenStreetMap.
- **SO7 – Centralized Data Fusion via AWS IoT Core**
  Establish secure data transmission to cloud using AWS IoT Core and PostgreSQL.
- **SO8 – Weighted Disaster Risk Prediction Modeling**
  Develop a machine learning model combining DEM, rainfall, and river data to predict flood risks.
- **SO9 – Full-Stack Administrative Dashboard Development**
  Build a centralized monitoring dashboard using Next.js.
- **SO10 – System Calibration & Prototype Validation**
  Assemble prototype, calibrate sensors, and validate against historical flood data.

Member 2 – Kashmira S S | IT22059772

Sub-Component: Human and Resource Demand Prediction

- Intelligent crowd detection and human counting using camera-based IoT and ML
- Demographic estimation for food, water, and medicine demand prediction
- Edge-cloud hybrid model for real-time and future forecasting
- Integration with cloud-based dashboard
- Comparison of AI-based predictions with manual estimation methods
- Fingerprint-based pre-disaster community registration and in-disaster verification of identities using R307 biometric scanner modules
- Wristband barcode issuance system along with a scanning pipeline to distribute wristbands to users in a way that prevents wristband duplication and keeps track of entitlements
- Capturing demographic and medical need data during fingerprint registration to enable structured estimation of resource demand
- Edge-cloud hybrid resource demand prediction model using profile data of registered users and historical consumption data
- MQTT-DR protocol design: lightweight extension of MQTT protocol using a 4-byte header, single-byte topic encoding, and binary compression of payloads using MessagePack
- Modeling network degradation to predict network quality 60 seconds into the future using a Random Forest classifier to adjust MQTT QoS levels proactively
- Implementation of SAAMP framework: Source-Action Classification Schema, Network State Monitor, Adaptive QoS Decision Engine, and Priority Buffer with CRITICAL-first reconnection flushing
- Integration of all modules including admin dashboard to display human count in real-time along with distribution status and prediction data
- Comparison of AI-based resource demand prediction against traditional manual estimation: assessed using MAE and RMSE metrics
- MQTT-DR protocol performance comparison against traditional MQTT protocol in three network simulation scenarios: critical message delivery rate, latency, network efficiency, data loss rate

Member 3 – Deerasinghe A D S N S | IT22065780

Sub-Component: Psycho-Attentional Gated Education System

- Design gamified Go/No-Go task (Tap the Animal – 90 seconds)
- Calculate Cronbach's alpha for reliability
- Implement binary gate decision logic (threshold 55/100)
- Implement calming breathing activity
- Re-assessment flow after failure

- Second gate decision and fail logic
- Implement `consecutive_low_scores` tracking in SQLite
- Perform chi-square analysis on pass rates
- Calculate session recovery rates
- Design Grade 3 & 4 curriculum-based activities
- Submit content for ZEO Homagama validation
- Develop content delivery interface with tracking
- Calculate curriculum alignment rating
- Perform linear regression on performance data
- Implement Z-score computation
- Implement alert trigger logic
- Develop Isolation Forest model (AWS Lambda)
- Deploy Next.js frontend & Node.js backend on Raspberry Pi
- Configure SQLite (WAL mode)
- Implement sync queue and service
- Connectivity detection & auto-sync
- Conduct offline simulation testing
- Perform load testing (Apache JMeter)
- Conduct failure testing (power interruption)
- Validate data consistency between SQLite and PostgreSQL

Timeline Alignment

Phase	Activity	Semester	Weeks
Phase 1	Topic selection, literature review, proposal writing	Semester 1	W1 – W11
Phase 2	System architecture & component design	Semester 1	W10 – W18
Phase 3	Hardware procurement & Smart Disaster Box assembly	Semester 1	W14 – W20
Phase 4	Sub-component development (parallel)	Semester 1–2	W18 – W32
Phase 5	System integration & testing	Semester 2	W30 – W38
Phase 6	Validation, testing, refinement	Semester 2	W36 – W42
Phase 7	Final report, paper, viva preparation	Semester 2	W38 – W48

Balanced Workload Distribution

- **Deerasinghe A D S N S**
    - Anomaly detection
    - Welfare alert system
    - Smart Camp Box hardware
    - Documentation
- **Hansana G P**
    - Environmental risk prediction
    - GPS altitude modeling
    - Multi-camp dashboard
    - Hardware integration
    - Cloud setup
- **Kashmira S S**
    - Human detection and crowd counting
    - Resource demand forecasting
    - MQTT message prioritizing base on network availability
    - Dashboard data integration

Shared Responsibilities (All Members)

- Joint system architecture design
- Co-authoring research paper and viva presentation
- Smart Disaster Box hardware integration
- System-level integration testing

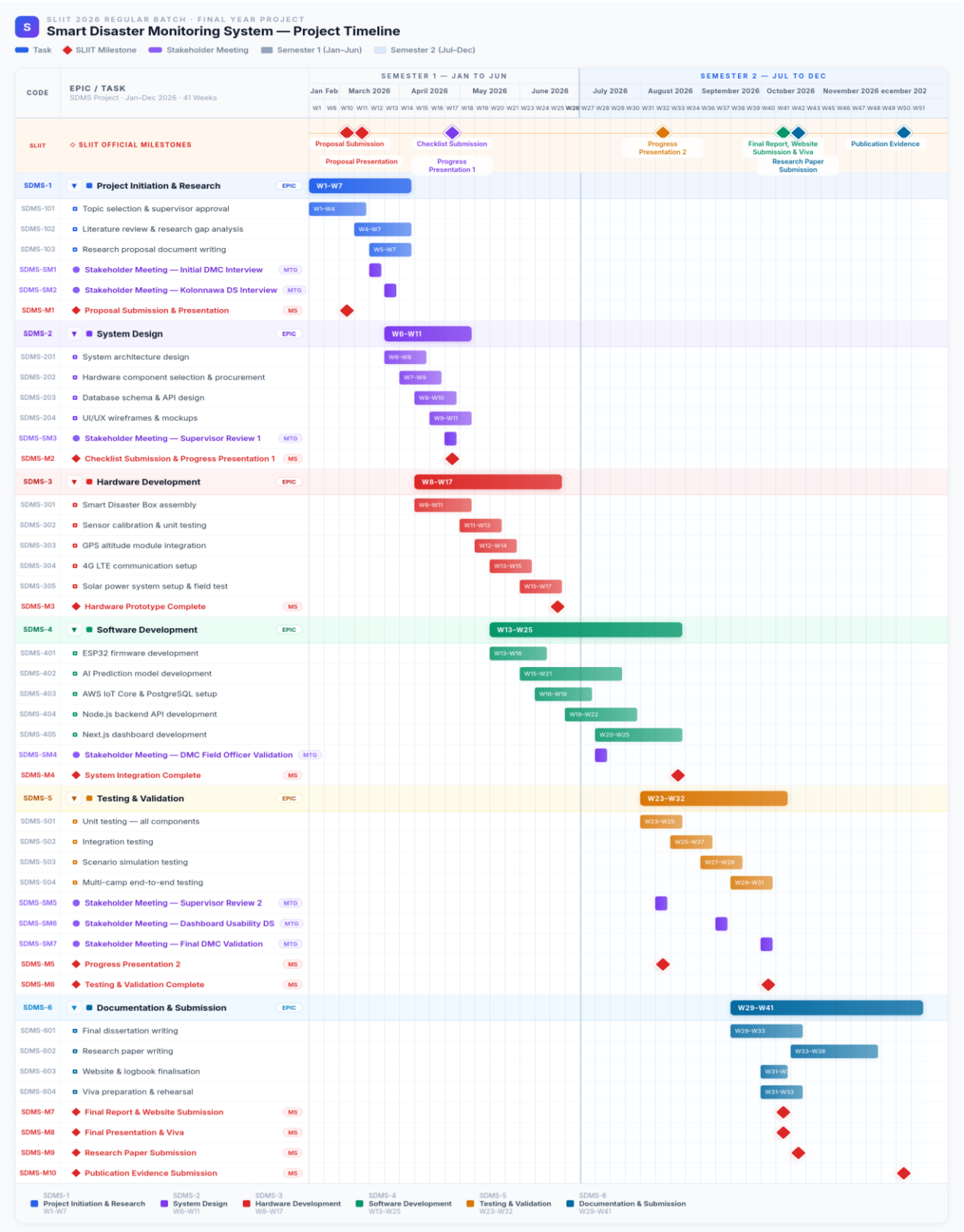
Collaborative Technical Contributions

Since the Smart Camp Box operates in **low-connectivity and low-power environments**, all members jointly contribute to:

- **Low Network Synchronization**
  Implement delayed and selective data syncing when connectivity is restored
- **Hop-Based Database Model**
  Design lightweight storage to ensure minimal footprint and high data integrity

- **Centralized Dashboard Integration**
  Develop a unified dashboard using Next.js and Node.js to aggregate all sub-components

9. GANTT CHART



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11.APPENDICES

Appendix A: Survey Instruments

The survey instruments are designed to collect qualitative and quantitative data from potential stakeholders, including disaster management officials and volunteers.

Sample Survey Sections:

- User experience with current disaster management systems
- Challenges in identity verification and resource distribution
- Importance of offline system capabilities
- Feedback on biometric and wristband-based identification
- Expectations from IoT-based disaster management solutions

Data Collection Method:

- Online questionnaires (Google Forms)
- Structured interviews with domain experts

Appendix B: Detailed System Diagrams

This appendix includes detailed diagrams representing the system design and workflows.

Included Diagrams:

- Component Diagram of Smart Camp Box
- Sequence Diagram for Message Flow
- Data Flow Diagram (DFD)
- Deployment Diagram (Edge + Cloud Architecture)
- Network Architecture Diagram

These diagrams provide a deeper understanding of system interactions beyond the high-level architecture.

Appendix C: Preliminary UI Mockups

Preliminary UI designs are created to visualize user interaction with the system.

Key Interfaces:

- Registration Interface (Fingerprint + User Details)
- Admin Dashboard (Resource Monitoring & Alerts)
- Distribution Interface (Barcode Scanning)
- Awareness Content Portal (Mobile View)

Design Tools Used:

- Figma

These mockups focus on usability, multilingual support, and simplicity for non-technical users.

Refer here: figma.com/make/iKK5Nqm5JCpvORpCbKnvev/Disaster-Relief-Camp-Management-UI?t=0lH2IoxTyy2BCjD0-20&fullscreen=1&preview-route=%2Fapp%2Fdashboard

Appendix D: Risk Analysis

A comprehensive risk assessment was conducted to identify potential challenges.

Risk	Impact	Mitigation Strategy
Hardware failure	High	Use backup devices and redundancy
Network failure	High	Offline-first architecture
Data loss	High	Local storage + periodic sync
Biometric errors	Medium	Multi-verification methods
User resistance	Medium	Training and awareness
Security threats	High	Encryption and secure storage

Appendix E: Ethical Documents

This research adheres to ethical standards in data collection and system implementation.

Key Ethical Considerations:

- Informed consent from all participants
- No storage of raw biometric data
- Use of non-reversible fingerprint templates
- Data minimization principles

- Confidentiality and anonymity of participants

Documents Included:

- Consent Forms
- Ethics Approval Letter (if available)
- Data Protection Policy Statement